

NAVRAJ SINGH KAMBO

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EDUCATION

University of British Columbia, Vancouver Master of Engineering (MEng), Electrical & Computer Engineering <i>Embedded Systems & Power Electronics</i>	<i>September 2026 - Present</i> Overall GPA: 85%
British Columbia Institute of Technology, Burnaby Bachelor of Engineering (BEng), Electrical <i>Graduated with Distinction</i>	<i>September 2015 - May 2019</i> Overall GPA: 85%

SCHOLARSHIPS AND AWARDS

- 2019 BCIT Student Innovation Challenge: 2nd place
- 2018 NSERC Undergraduate Student Research Award (USRA) recipient
- 2017 APEGBC Scholarship recipient

RESEARCH & CO-OPS

British Columbia Institute of Technology, Burnaby <i>SoE / Senior Capstone Project - Autonomous Vineyard Harvesting</i>	<i>September 2018 - May 2019</i>
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- Developed machine vision pipeline for real-time detection of wine grape clusters using classical image processing and machine learning
- Designed a geometric mapping algorithm to infer vineyard boundaries using vine posts
- Implemented serial communication protocols to coordinate navigation, vision, and control subsystems
- Created an autonomous robotic platform for precision vineyard navigation and harvesting

British Columbia Institute of Technology, Burnaby <i>Research Intern</i>	<i>May 2018 - August 2018</i>
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- Concluded that concrete cylinders have only one reliable striking location for strength estimation
- Developed a cross-platform mobile strength estimation app (iOS & Android) for use in resource-constrained environments with concrete bricks
- Processed historical sensor data and implemented signal-processing algorithms to automate peak frequency detection for impact analysis
- Implemented and debugged time-domain impulse identification feature for mobile devices

Wireless Technical Services Inc., Coquitlam <i>Co-op Student</i>	<i>April 2017 - August 2017</i>
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- Supported security and QA efforts on Fortis BC Radio Network by identifying vulnerabilities using Software Defined Radio (SDR)
- Wrote Python automation scripts for configuration backups and diagnostics on Cisco routers and switches
- Performed QA testing including Rack Acceptance Tests (RATs) for network infrastructure deployments

PROFESSIONAL DEVELOPMENT

Agritech 4.0 Bootcamp (MECH420) - Simon Fraser University, SFU MSE <i>Dr. F. Najafi, R. Raksha, A. Marzouk, K. Vijayaraghavan, B. Lu, M. Narimani, T. Al-Khudairi</i>	<i>2025</i>
Converter Control (ECEA 5702) - University of Colorado Boulder, Coursera <i>Dr. Robert Erickson</i>	<i>2025</i>

Converter Circuits (ECEA 5701) - University of Colorado Boulder ↗, Coursera <i>Dr. Robert Erickson</i>	2025
Embedded C Programming Design Patterns ↗, Udemy <i>Martin Schröder</i>	2024

TECHNICAL SKILLS

Embedded Systems: C/C++, RTOS, CMAKE, OpenOCD, GDB, STM32, Renesas, TI SYS/BIOS,

Electronics Debugging & Analysis: Analog/digital circuit analysis, PCBA Design-for-Test (DFT), oscilloscopes, logic analyzers, signal tracing, RS-485, CAN, USB, UART/I2C/SPI, SWD

Mechanical Design: CAD, CAM, CNC fabrication workflows, mechanical part and test fixture design, SolidWorks, HSMXpress, 3D printing

Modeling & Simulation: MATLAB, Simulink, LTSpice, ModelSim

Software & Tools: Git, JetBrains CLion/PyCharm, Jira, MS Office

Languages: C/C++, Python, System Verilog, JavaScript, L^AT_EX

PROFESSIONAL EXPERIENCE

Delta Intelligent Building Technologies (Canada) Inc. ↗, Surrey *April 2023 - Present*
Test & Automation Engineer EIT

- Develop embedded test firmware (C/C++) and Python-based sensor drivers for automated testing
- Review Altium PCB layouts for DFM/DFT compliance and contribute to hardware design reviews
- Design and validate custom bed-of-nails test fixtures in SolidWorks for high-throughput PCBA validation, and utilize fixtures for board bring-up activities
- Collaborate on test plan development and process improvement within proprietary QA framework
- Support production workflow optimization through cross-team collaboration and test result analysis

Wellons Canada ↗, Surrey *April 2022 - April 2023*
Electrical Engineer EIT

- Drafted construction documentation including P&IDs for industrial biomass and gas-fired energy systems
- Diagnosed and debugged Rockwell PLC and HMI programs, enhancing reliability of kiln control systems
- Designed MCC single-line diagrams, control schematics, and CSA B149.3:20 compliant gas trains
- Supported onsite commissioning at lumber mills across BC and Alberta, validating electrical and control systems under field conditions

Apex Motion Control, Inc. ↗, Surrey *August 2019 - April 2022*
Design Engineer EIT

- Led project lifecycle for collaborative robotic systems in the food and beverage industry, from design through deployment
- Programmed robotic decorating ↗ and pick-and-place ↗ motion sequences, optimizing quality and cycle-time
- Drafted CEC-compliant electrical schematics and wired three-phase electrical panels
- Created mechanical designs for robotic frames, and hygienic conveyor systems ↗ in SolidWorks, meeting customer-specific requirements

BKS Cablecom Ltd. ↗, Burnaby *May 2016 - August 2016*
Jr. Technician

- Installed and tested data cabling for data and voice systems

- Labeled and assembled patch panel assemblies, and upgraded wireless networking equipment
- Assisted senior technicians and IT teams in network upgrades at UBC and enterprise client sites
- Applied TIA/EIA cabling standards and industry best practices for telecom installations

PRESENTATIONS

- BCIT BEng Electrical Speakers Night (February 10th 2025)
- White Rock Christian Academy Career Fair 2025 (January 27th 2025)
- BCIT BEng Electrical Speakers Night (February 12th 2024)
- BCIT BEng Electrical Speakers Night (February 8th 2023)
- BCIT BEng Electrical Speakers Night (February 2nd 2022)
- BCIT ELEX 7825 Guest Lecture (November 22nd 2021)
- Brookwood Secondary School Science 10 Robotics Virtual Seminar (December 4th 2020)

VOLUNTEER WORK

Guru Nanak's Free Kitchen , East Vancouver *Volunteer*

September 2013 - Present

- Provide weekly meal service to under-served communities in Vancouver's Downtown Eastside
- Assist with food prep, cooking, and sanitation to support large-scale community outreach
- Collaborate with volunteers to ensure efficient setup, distribution, and clean-up operations
- Occasionally coordinate volunteer tasks and help onboard new team members
- Partner with university and community volunteer groups, including UBC student organizations , to strengthen engagement and broaden outreach impact

EXTRA-CURRICULAR INTERESTS

- Playing soccer at a competitive level (Fraser Valley Soccer league, Mens Divison 1)
- Running and hiking
- Road and mountain biking between Burnaby and Vancouver
- Practicing martial arts (Hapkido - Blue Belt)
- Seasonal snowboarding
- 3D printing, digital art, and drawing